Design, Fabrication, and Analysis of a Mach-Zehnder Interferometer on a Silicon-on-Insulator Platform

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***Abstract*—This paper presents the design, modeling, fabrication, and data analysis of a Mach-Zehnder interferometer (MZIs) on a silicon photonic platform. The design objective is to observe the effect of the path length variation on the spectral response of an interferometer circuit. Different MZI configurations are designed using KLayout which is then sent for fabrication using electronic beam lithography. The optical transmission results would be then compared with the simulations on Lumerical INTERCONNECT to verify the validity of the fabricated design.**

***Index Terms*—Mach-Zehnder interferometer, fabrication, silicon-on-insulator, electronic beam lithography**

I. Introduction

The growing field of photonic integrated circuits (PICs) has gained attention to address the limitations of electronics in terms of scalability and computational power. By using light as the carrier of information, different properties of light such as wavelength, polarization, and mode, can be manipulated for signal processing.

One of the simplest circuit topologies in silicon photonics is the Mach–Zehnder interferometer (MZI). An MZI consists of a beam splitter at the input port and a beam combiner at the output port, connected by two optical waveguide arms. Depending on the relative phase difference between these two paths, the optical signals interfere either constructively or destructively at the output. Variations in the path length of one arm introduce a phase shift, which directly influences the interference condition and, consequently, the output intensity of the device.

II. Theory of Operation

A typical beam splitter on an MZI has a splitting ratio of 50:50, in which light travels in two different waveguide arms. If there is a difference in the path length which would induce a phase difference, then there would be a noticeable shift of interference pattern at the output. This is governed by equation (1):

$$I_{out} = I_0 \cos^2 (\frac{\Delta\theta}{2}) \tag{1}$$

This phase difference $\Delta\theta$ is related to the path length difference ΔL and the effective index n_{eff} using equation (2)

$$\Delta\theta = (\frac{2\pi n_{eff}\Delta L}{\lambda}) \tag{2}$$

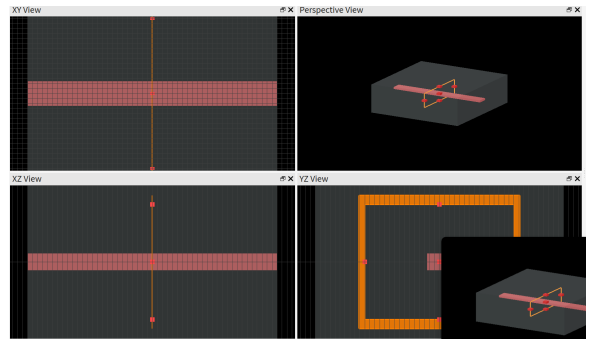

Fig. 1. Simulation setup for Lumerical MODE.

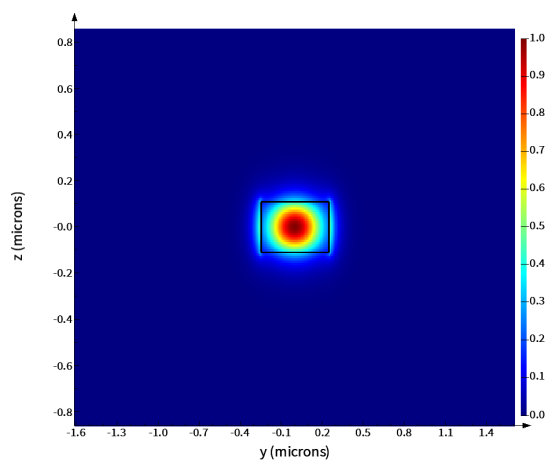

Fig. 2. Mode profile of the waveguide.

III. Modeling and Simulation

This design uses a silicon-on-insulator platform, with silicon dioxide (SiO_2) as the cladding and silicon as the waveguide material. In order to confirm the behavior of the light in a waveguide, simulations are performed using Lumerical MODE, with the setup illustrated in Figure 1. For a given width of $0.5\mu m$ and a thickness of $0.22nm$, the mode profiles for the fundamental TE mode is shown in Figure 2.

Using a script in Lumerical MODE, the compact model of this waveguide at a wavelength of $1.55\mu m$is given by equation (3).

$$n_{eff}(\lambda) = 2.445 - 1.143*(\lambda - 1.55) - 0.4370*(\lambda - 1.55)^2 \tag{3}$$

Moving forward, the MZI is simulated in Lumerical INTERCONNECT to visualize the transmission spectrum. Figure 3 shows the schematic set-up of the MZI simulation, consisting

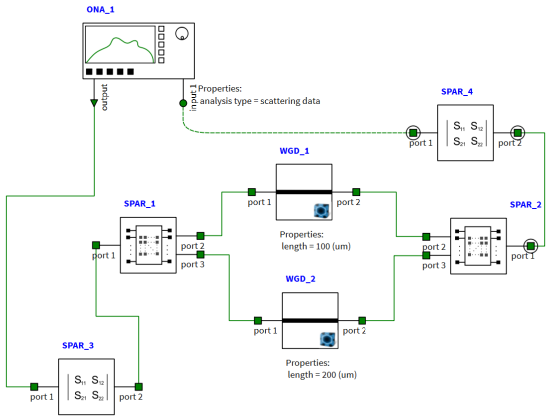

Fig. 3. Mach-Zehnder interferometer in Lumerical INTERCONNECT

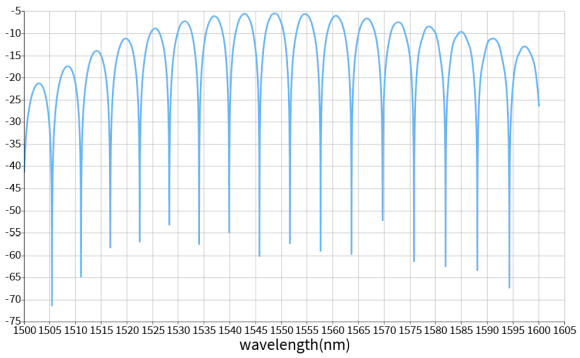

Fig. 4. Transmission in dB vs Wavelength of an MZI with $\Delta L = 100\mu m$

of an optical network analyzer as the visualizer, connected to a grating coupler modeled using its S-parameters.

In this simulation, Y-branch was used as the beam splitters, and a waveguide of two different lengths (100 μm and 200 μm) are connected in between them. After running, the transmission spectrum is generated by the optical network analyzer, as presented in Figure 4.

To physically implement this interferometer, the KLayout software is used to design and generate the GDS files that will be sent to the fabrication facility. Figure 5 illustrates the mask created in KLayout. This consists of five MZIs with different path lengths and a loopback structure for calibration, which is typically a pair of grating couplers connected by a waveguide.

IV. Fabrication and Experimental Results

The fabrication of the chip is to performed at the University of British Columbia's 100keV electron beam lithography by Applied Nanotools Inc., Canada.

Measurements are obtained using the automated optical probe station of the university. The transmission spectrums for each MZI with different path lengths are shown in Figures 6 to 10.

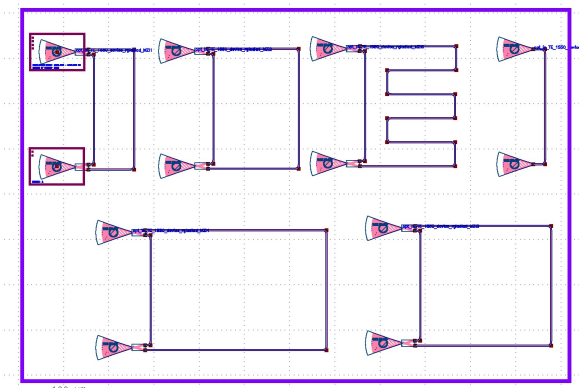

Fig. 5. MZI schematic created in KLayout using Y-branch with five different path length difference, right to left and top to bottom: $\Delta L = 100\mu m, 200\mu m, 500\mu m$, a loopback calibration structure, and $\Delta L = 400\mu m, 300\mu m$.

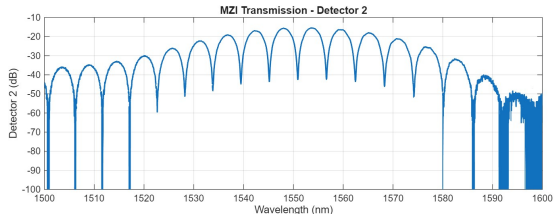

Fig. 6. Experimental transmission spectrum of a Mach–Zehnder interferometer (MZI) with a path length difference of $\Delta L = 100\,\mu$m.

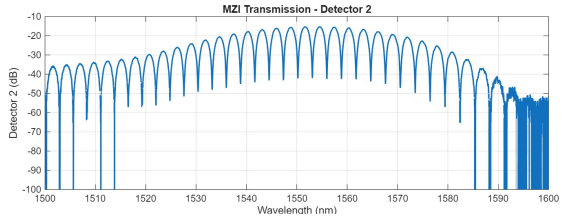

Fig. 7. Experimental transmission spectrum of a Mach–Zehnder interferometer (MZI) with a path length difference of $\Delta L = 200\,\mu$m.

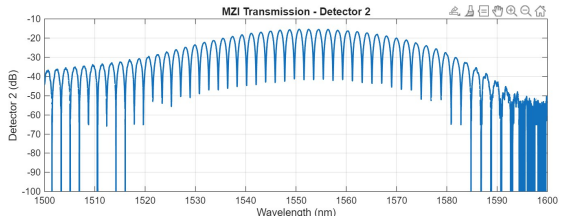

Fig. 8. Experimental transmission spectrum of a Mach–Zehnder interferometer (MZI) with a path length difference of $\Delta L = 300\,\mu$m.

V. Analysis

The free spectral range (FSR) is a key parameter in interferometric devices such as the Mach–Zehnder interferometer (MZI). It is defined as the wavelength spacing between consecutive transmission minima or maxima in the interference spectrum. The FSR is inversely proportional to the optical path length difference (ΔL) between the two arms of the interferometer, which means that larger path differences would make the adjacent fringes closer to each other. This is observed

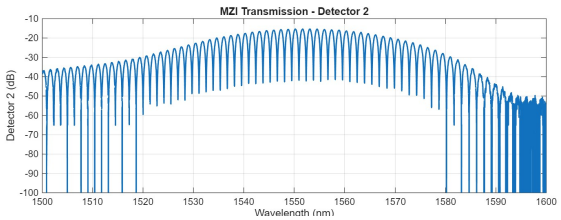

Fig. 9. Experimental transmission spectrum of a Mach–Zehnder interferometer (MZI) with a path length difference of $\Delta L = 400\,\mu$m.

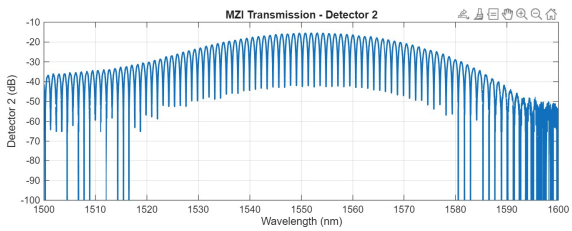

Fig. 10. Experimental transmission spectrum of a Mach–Zehnder interferometer (MZI) with a path length difference of $\Delta L = 500\,\mu$m.

in the figures of the transmission spectrum in the previous section, verifying that our devices behave as intended.

Using the computed value from the FSR, the group index for each device can be calculated using Equation 4:

$$n_g = \frac{\lambda^2}{\mathrm{FSR} \cdot \Delta L} \tag{4}$$

Table I displays the corresponding FSR and group index values for each MZI device.

TABLE I
MEASURED FSR, PATH LENGTH DIFFERENCE, AND CALCULATED GROUP INDEX

ΔL (μm)	FSR (nm)	n_g
99.884	5.73	4.1978
199.884	2.85	4.2085
299.884	1.91	4.1813
399.884	1.43	4.2014
499.884	1.15	4.1792

VI. Design Variability Analysis using Monte Carlo Simulations

The fabrication of actual silicon photonic integrated circuits faces challenges in variability, which impacts circuit performance. Differences in the waveguide thickness, feature sizes, and losses in the optical testing set-up contributes to the deviation of the experimental results with the simulation.

One way to model these manufacturing variability is to conduct corner analysis and Monte Carlo simulations. The SiEPIC-EBeam-PDK has a built-in function to conduct Monte Carlo simulations by integrating KLayout and Lumerical INTERCONNECT.

For simplicity, the chosen device for this simulation is the MZI with a path length difference of $100\mu m$ ($99.884\mu m$ to be exact). The waveguide should be 220 nm thick and 500 nm wide.The fabrication technology is specified to *Electron Beam Lithography (Applied Nanotools)* and simulations are conducted on five test wafers and five test die for wafers. The histogram results for the FSR are displayed in Figure 11, which is also tabulated in Table II.

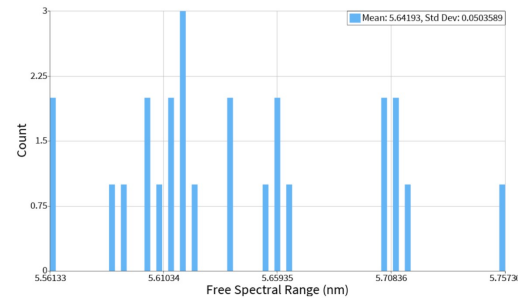

Fig. 11. Histogram of the Monte Carlo simulations conducted in an MZI device with $\Delta L = 100\mu m$.

TABLE II
MEASURED FSR, PATH LENGTH DIFFERENCE, AND CALCULATED GROUP INDEX

FSR (nm)	n_g
5.56261	4.3240
5.58807	4.3043
5.60334	4.2926
5.64193	4.2632
5.71027	4.2122
5.71536	4.2085
5.75609	4.1786

From this, we can conclude that the expected range of group index values is from 4.1786 to 4.3240. Going back to Table I, all computed group index values lie within the range from our Monte Carlo simulations. Thus, we have successfully validated and modeled the variability of our fabricated circuit in comparison with the simulations.

VII. Conclusion

In this work, a Mach–Zehnder interferometer was successfully designed, fabricated, and experimentally characterized on a silicon-on-insulator platform. The measured transmission spectra showed the expected interference behavior, where increasing the path length difference ΔL resulted in a smaller free spectral range (FSR). Using the extracted FSR values, the group index for each device was calculated and found to be consistent across all designs. These values also fall within the range predicted by the Monte Carlo simulations, which shows good agreement between simulation and experimental results. Overall, this validates both the design approach and the fabrication process, and demonstrates the reliability of the MZI structure for use in silicon photonic circuits.

References

[1] L. Chrostowski, "Silicon photonics design, fabrication and data analysis." https://learning.edx.org/course/course-v1:UBCx+Phot1x+1T2025/home, 2026. edX Course, University of British Columbia.

[2] L. Chrostowski and M. Hochberg, *Silicon Photonics Design.* Cambridge University Press, 2015.